Peer Review: "Multi Agent Systems" (Jakub Janicek, Oscar Schuyl, Matei Tanasa, Archisha Chanda)

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Summary

This paper looks at how different network topologies affect the success of manipulative agents in the HK model. The authors compare Erdős-Rényi, Watts-Strogatz, and Barabási-Albert networks and measure how much manipulators can shift the opinions of normal agents.

I think the topic is interesting and the paper has a clear research question. However, there are some issues with the experimental setup and results section that make the conclusions a bit harder to evaluate.

Recommendation: Revision

1. Is it clear what challenge the authors are tackling?

Yes. I found the introduction easy to follow, and the research question is clearly stated. The motivation for studying manipulative agents on different network structures also makes sense.

One thing I noticed is that the abstract makes the difference between the network types sound quite large, while the actual results seem fairly close together. I also noticed that Section 1.2 mentions a fully connected network as a comparison, but I could not find results for it later in the paper.

2. Is the model appropriate for the challenge?

Overall, I think so. Restricting interactions to network neighbours seems like a natural extension of the original model, and keeping the same agent types helps isolate the effect of topology.

My main concern is that topology and density seem to be mixed together in the experiments. Since the paper itself shows that density affects manipulation success, it is difficult to know how much of the result comes from topology alone. I think matching the networks on density would make the comparison stronger.

3. Is the model described in enough detail to implement it?

Not completely. I understand the general model, but there are some details that I would still need before implementing it myself.

For example, the initial opinion distributions are not clearly described, and there are some inconsistencies in the reported parameter ranges and number of runs. The value of ϵ should also be stated more clearly in the methods section instead of only appearing in figure captions.

4. Are similarities to existing literature clearly indicated?

Yes. The related work section is well organized and makes it clear how this paper builds on previous studies.

There are just a few small things that could be improved. The original HK paper should probably be cited, and the description of stubborn agents in Section 2 appears to contain a mistake. Some comparisons with previous work could also be explained a bit more carefully since not all studies measure the same outcomes.

5. Are results presented in a way that is easy to understand?

This is probably the section that needs the most work. Some figures still seem unfinished, with placeholder text remaining and Figure 6 missing a caption.

I also found it difficult to judge how meaningful some of the differences are because no error bars or confidence intervals are shown. It would help if the statistical tests were explained in a bit more detail, especially what the reported observations actually represent.

6. Do the conclusions follow from the results?

For the most part, yes. I think the conclusion is one of the stronger parts of the paper because it discusses the limitations and does not overstate the findings.

That said, a few claims elsewhere in the paper feel slightly stronger than what the results show. In particular, the explanation involving hubs in Barabási-Albert networks sounds reasonable, but it is not directly tested in the paper.

Minor Issues and Typos

There are several spelling and grammar mistakes throughout the report, such as "nework", "newtork", and "startegy". I also noticed some inconsistent capitalization and naming of network types. A careful proofreading pass would improve readability.

Overall Recommendation

I think this is an interesting paper with a clear research question and a good overall idea. The biggest issue for me is that the effect of topology is difficult to separate from the effect of density in the current experiments. Clarifying some methodological details and improving the presentation of the results would make the paper much stronger.